

Prysm EIP-8025 update

Optional proofs

2026-05-13



PRYSM
By Offchain Labs

1 - EL + signed proof generation

```
1 vc_extra_params:
2   - --generate-proofs
3   - --prover-rest-api-provider=http://zkboost-1:3000
```

1. BN gets block from gossip
2. VC **pulls the block** from beacon node, generates NewPayloadRequest and sends it to ZK boost
3. ZK boost **pulls the witness** from EL, generates the proof and sends it to VC
4. VC **signs the proof** and sends it to BN
5. BN **gossips** the proofs

2 - EL + execution AND verification

```
1 cl_extra_params:
2   - --zkvm=execute-and-verify
3   - --verifier-rest-api-provider=http://zkboost-2:3000
```

1. BN **gets** block + proofs from gossip
2. BN **sends** the exec. payload to EL **AND sends** proofs to ZK boost
3. EL **executes** the payload **AND** ZK boost **verifies** the proofs
4. If ok, block is imported **normally**, if not, **optimistically**
5. (Missing proofs can be eventually retrieved via RPC)

3 - NO EL - verification only

```
1 - el_type: None
2 ...
3 cl_extra_params:
4   - --zkvm=verify-only
5   - --verifier-rest-api-provider=http://zkboost-2:3000
```

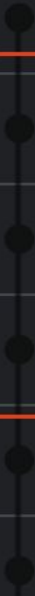
- 1. BN **gets** block + proofs from gossip
- 2. BN **sends** proofs to ZK boost
- 3. ZK boost **verifies** the proofs
- 4. If ok, block is imported **normally**, if not, **optimistically**

4 - Block proposed with BN without an EL (builder mandatory)



Most recent slots

[View more](#)

Chain	Epoch	Slot	Status	Time	Proposer
	3	122	Proposed	7 sec. ago	 3-reth-prysm
	3	121	Proposed	19 sec. ago	 4-prysm
	3	120	Proposed	31 sec. ago	 4-prysm
	3	119	Proposed	43 sec. ago	 4-prysm
	3	118	Proposed	55 sec. ago	 1-reth-lighthouse
	3	117	Proposed	1 min. ago	 1-reth-lighthouse

TODO:

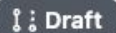
- Rebase the **optional-proof** branch on the main **develop** branch (lot of conflicts now)
- Clean the code (right now it's not releasable quality)
- Add unit tests
- Optional proofs are based on Fulu $\Rightarrow$ Move to Gloas?

What about a devnet?

- We have 2 Kurtosis interop-able implementations (LH + Prysm)
- Should the devnet be based on Fulu or Gloas?

<https://github.com/OffchainLabs/prysm/pull/16352>

Optional execution proofs #16352

 Draft nalepae wants to merge 88 commits into develop from optional-proofs

 Conversation 6  Commits 88  Checks 7  Files changed 147



nalepae commented on Feb 12 • edited

Member

What type of PR is this?

Feature

Some context:

What does this PR do? Why is it needed?

Continuation of:

-  [Rough prototype for architectural changes needed to introduce execution proofs #16003](#)

How to run it:

The Lighthouse image must be built from:

- <https://github.com/eth-act/lighthouse/tree/optional-proofs>

with the following command:

```
docker build -t lighthouse:latest .
```

Warning

Currently this PR is based on Fulu.

As a consequence, if a Prysm beacon node not connected to any EL (`el_type: None`) has to propose a block and if the MEV-boost instance does not return any header, then the block is missed.

It will be naturally solved when switching to Gloas

Prover: Lighthouse

```
participants:
# Prover node
- el_type: reth
```